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Data Analytics with R

Quality Laboratory Manual

Experiment No. 02

To install R and RStudio and create vector, matrices, arrays and factors and perform basic arithmetic operations and logical comparisons of R objects



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Experiment No. 02

Title of Experiment: To install R and RStudio and create vectors, matrices, arrays and factors and perform basic arithmetic operations and logical comparisons on R objects

Aim of Experiment: To install and configure the R programming environment and to understand fundamental R data structures such as vectors, matrices, arrays and factors along with performing arithmetic and logical operations on R objects

System Requirements –

- Operating System: Windows 10 or above
- RAM: Minimum 4 GB
- Hard disk: 10 GB free space
- Processor: 2.33 GHz or higher

Software/s Needed for Experiment

- R programming language (latest stable version)
- RStudio IDE
- Spreadsheet software (Microsoft Excel/ Libre Office)
- Internet Connection (for dataset access)

Experiment Objectives:

- To install and configure R and RStudio successfully
- To understand the RStudio working environment
- To create and manipulate vectors in R
- To construct matrices and arrays for multidimensional data
- To use factors to represent categorical data
- To perform arithmetic operations on R objects
- To apply logical comparisons and relational operators in R

Experiment Outcomes:

After completing the experiment, students will be able to –

- Work confidently in the RStudio environment
- Create and manipulate fundamental R data structures
- Perform numerical computations using R
- Apply logical operations for data filtering and comparison
- Build foundational programming skills required for data analytics

Theory:

R is an open-source programming language widely used for statistical computing and data analytics. It provides powerful data structures that allow efficient data manipulation and analysis.

1. Installation of R and RStudio

R is the core programming engine, while RStudio provides an integrated development environment (IDE) that simplifies coding, visualization, and project management.

2. R Data Structures

a) Vectors

A vector is the simplest data structure in R that stores elements of the same data type.

Example:

```
x <- c(10, 20, 30, 40)
```

Types of vectors:

- Numeric
- Character
- Logical

b) Matrices

A matrix is a two-dimensional collection of elements arranged in rows and columns.

```
m <- matrix(1:9, nrow = 3, ncol = 3)
```

c) Arrays

Arrays extend matrices to more than two dimensions.

```
a <- array(1:12, dim = c(3,2,2))
```

d) Factors

Factors are used to represent categorical data.

```
f <- factor(c("Low", "Medium", "High"))
```

3. Arithmetic Operations in R

R supports element-wise arithmetic operations:

```
x + y
```

```
x - y
```

```
x * y
```

```
x / y
```

4. Logical Comparisons

Logical operators compare values and return TRUE or FALSE.

Operator	Meaning
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>	Greater than
<	Less than
==	Equal to
!=	Not equal
>=	Greater or equal
<=	Less or equal

Task to Perform

Case Study-Based Observation Task – Student Performance Analysis Using Basic R Data Structure

A college wants to analyze the academic performance of student enrolled in a data analytics course. The department has collected information about students including marks obtained in different subjects, attendance percentage and performance categories (Pass/ Fail/ Distinction)

Before performing advanced analytics, the data analyst must organize and manipulate the data using basic R data structures such as vectors, matrices, arrays and factors and perform arithmetic and logical operations to extract meaningful insights

You are assigned as a junior data analyst to perform the initial data preparation using R

Dataset Description –

The following information is collected for 5 students:

Student	Math	Statistics	Programming	Attendance (%)	Result
S1	78	82	75	90	Pass
S2	65	70	68	85	Pass
S3	92	88	95	96	Distinction
S4	55	60	58	72	Pass
S5	40	45	50	65	Fail

Task-1 –

Installation Verification

Open RStudio and verify successful installation using: `version`

Task-2 –

Create a vector

Create vectors for marks of each subject

```
math <- c(78, 65, 92, 55, 40)
stats <- c(82, 70, 88, 60, 45)
prog <- c(75, 68, 95, 58, 50)
```

Identify the type and length of vectors

Task-3 –

Create a Matrix

Combine subject marks into a matrix

```
marks_matrix <- matrix(c(math, stats, prog),
```

```
nrow = 5,  
ncol = 3)
```

Display the `marks_matrix` and identify rows and columns

Task-4 –

Perform Arithmetic Operations

Calculate total and average marks

```
total <- math + stats + prog  
average <- total/3
```

Observe and note element-wise operations

Task-5 –

Logical Comparisons

Find students scoring above 70 average marks

```
average > 70
```

Identify TRUE/ FALSE outputs

Task-6 –

Create an Array

Store marks across two semesters

```
semester_array <- array(1:30, dim = c(5,3,2))
```

Observe and note the multi-dimensional structure

Task-7 –

Create a Factor

Represent result categories using factors

```
result <- factor(c("Pass", "Pass", "Distinction", "Pass", "Fail"))
```

Check levels

Submission Guidelines

- 1) Include R code and outputs
- 2) Provide observation table
- 3) Answer reflection questions
- 4) Submit report (2-3 pages PDF)

Observations:

- R and RStudio were successfully installed and configured
- Vectors allowed efficient storage of homogeneous data
- Matrices organized data into tabular form suitable for mathematical operations

- Arrays supported multidimensional datasets
- Factors efficiently represented categorical variables
- Arithmetic operations were performed element-wise
- Logical comparisons returned Boolean outputs useful for decision-making

Conclusion:

The experiment successfully introduced the R programming environment and fundamental data structures used in data analytics. Students learned to create vectors, matrices, arrays and factors and performed arithmetic and logical operations. These concepts form the foundation for data manipulation and analysis in subsequent experiments

Expected Oral Questions:

- 1) What is R programming language?
- 2) Differentiate between R and RStudio
- 3) What is a vector in R?
- 4) How is a matrix different from an array?
- 5) What is a factor and where is it used?
- 6) Explain element-wise operations in R
- 7) Name logical operators in R
- 8) Why are factors important in data analytics?

FAQs in Interview:

- 1) Why R is preferred for data analytics?
- 2) What is the difference between vector and list?
- 3) Why are factors used instead of character data?
- 4) Are arithmetic operations applied element-wise in R?
- 5) Which data structure is most commonly used in R analytics?